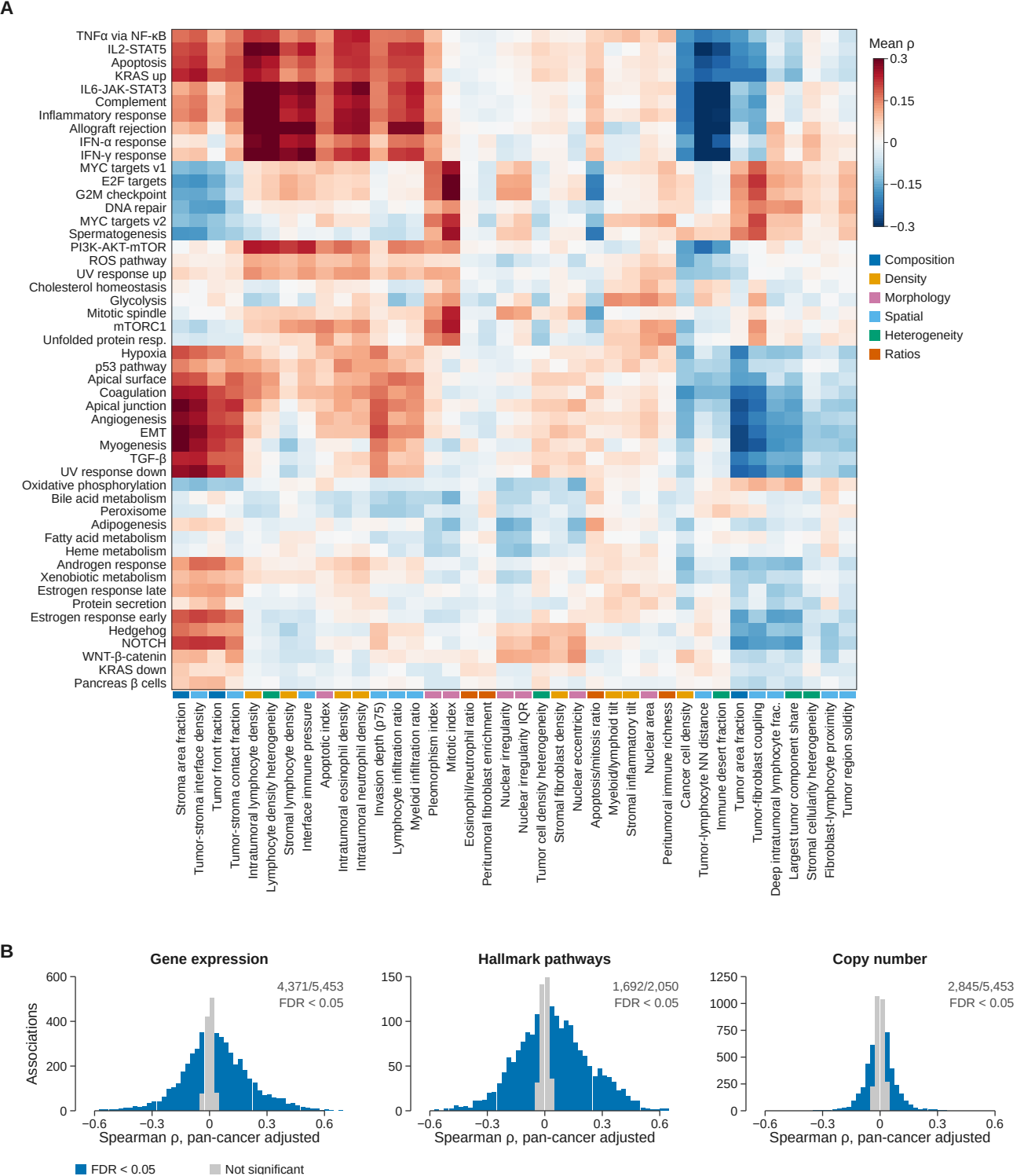
A
TNFα via NF-κB
IL2-STAT5
Apoptosis
KRAS up
IL6-JAK-STAT3
Complement
Inflammatory response
Allograft rejection
IFN-α response
IFN-γ response
MYC targets v1
E2F targets
G2M checkpoint
DNA repair
MYC targets v2
Spermatogenesis
PI3K-AKT-mTOR
ROS pathway
UV response up
Cholesterol homeostasis
Glycolysis
Mitotic spindle
mTORC1
Unfolded protein resp.
Hypoxia
p53 pathway
Apical surface
Coagulation
Apical junction
Angiogenesis
EMT
Myogenesis
TGF-β
UV response down
Oxidative phosphorylation
Bile acid metabolism
Peroxisome
Adipogenesis
Fatty acid metabolism
Heme metabolism
Androgen response
Xenobiotic metabolism
Estrogen response late
Protein secretion
Estrogen response early
Hedgehog
NOTCH
WNT-β-catenin
KRAS down
Pancreas β cells
Mean ρ
0.3
0.15
0
−0.15
−0.3
Composition
Density
Morphology
Spatial
Heterogeneity
Ratios
Stroma area fraction
Tumor-stroma interface density
Tumor front fraction
Tumor-stroma contact fraction
Intratumoral lymphocyte density
Lymphocyte density heterogeneity
Stromal lymphocyte density
Interface immune pressure
Apoptotic index
Intratumoral eosinophil density
Intratumoral neutrophil density
Invasion depth (p75)
Lymphocyte infiltration ratio
Myeloid infiltration ratio
Pleomorphism index
Mitotic index
Eosinophil/neutrophil ratio
Peritumoral fibroblast enrichment
Nuclear irregularity
Nuclear irregularity IQR
Tumor cell density heterogeneity
Stromal fibroblast density
Nuclear eccentricity
Apoptosis/mitosis ratio
Myeloid/lymphoid tilt
Stromal inflammatory tilt
Nuclear area
Peritumoral immune richness
Cancer cell density
Tumor-lymphocyte NN distance
Immune desert fraction
Tumor area fraction
Tumor-fibroblast coupling
Deep intratumoral lymphocyte frac.
Largest tumor component share
Stromal cellularity heterogeneity
Fibroblast-lymphocyte proximity
Tumor region solidity
B
Gene expression
4,371/5,453
FDR < 0.05
Hallmark pathways
1,692/2,050
FDR < 0.05
Copy number
2,845/5,453
FDR < 0.05
Associations
Spearman ρ, pan-cancer adjusted
FDR < 0.05
Not significant